

Government Expenditure in a Real Business-Cycle Model

Technology:

$$Y_t = \theta_t K_t^\alpha L_t^{1-\alpha} \quad \alpha \in (0, 1)$$

with

$$\ln \theta_{t+1} = (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1}$$

where $\psi \in (0, 1)$, $\theta > 0$, and $\{\varepsilon_{t+1}\}_{t=0}^{\infty}$ is a sequence of i.i.d. random variables with $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.

Government expenditure evolves according to

$$\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \eta_{t+1}$$

where $\phi \in (0, 1)$, $G > 0$, and $\{\eta_{t+1}\}_{t=0}^{\infty}$ is a sequence of i.i.d. random variables with $\eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$.

The shocks to technology and government expenditure are independent from each other.

The government's budget constraint in period t is

$$G_t = T_t$$

where T_t denotes revenues from lump-sum taxes.

The utility function of the representative agent is

$$U = \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - L_t)$$

with

$$u(C_t, 1 - L_t) = \ln C_t + \gamma \ln(1 - L_t)$$

where $\gamma > 0$ and $\beta \equiv \frac{1}{1+\rho}$ with $\rho > 0$.

The budget constraint of the representative agent is

$$C_t + I_t = w_t L_t + r_t K_t + \Pi_t - T_t$$

where

$$I_t = K_{t+1} - (1 - \delta)K_t$$

and Π_t are profits (that will be zero in equilibrium).

Profit Maximization

$$\Pi_t \equiv \max_{K_t, L_t} \theta_t K_t^\alpha L_t^{1-\alpha} - w_t L_t - r_t K_t$$

FOC

$$K_t : r_t = \alpha \theta_t K_t^{\alpha-1} L_t^{1-\alpha} = \alpha \frac{\theta_t K_t^\alpha L_t^{1-\alpha}}{K_t}$$

$$L_t : w_t = (1 - \alpha) \theta_t K_t^\alpha L_t^{-\alpha} = (1 - \alpha) \frac{\theta_t K_t^\alpha L_t^{1-\alpha}}{L_t}$$

Then (d stands for demand):

$$r_t = \alpha \theta_t \left(\frac{K_t}{L_t} \right)^{\alpha-1} = \alpha \frac{Y_t}{K_t^d}$$

$$w_t = (1 - \alpha) \theta_t \left(\frac{K_t}{L_t} \right)^\alpha = (1 - \alpha) \frac{Y_t}{L_t^d}$$

and

$$\Pi_t = 0$$

Notice that the real wage equals the marginal product of labor, which is proportional to the average productivity of labor $\frac{Y_t}{L_t}$.

Utility Maximization

$$\max_{\{C_t, L_t, K_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\ln C_t + \gamma \ln(1 - L_t)]$$

$$\text{s.t.} \quad C_t + K_{t+1} - (1 - \delta)K_t = w_t L_t + r_t K_t - T_t$$

$$K_0 > 0 \text{ given}$$

Lagrangian

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \left\{ \begin{array}{l} \beta^t [\ln C_t + \gamma \ln(1 - L_t)] + \\ \lambda_t [w_t L_t + r_t K_t + (1 - \delta)K_t - T_t - C_t - K_{t+1}] \end{array} \right\}$$

FOC (s stands for supply):

$$C_t : \quad \frac{\beta^t}{C_t} = \lambda_t$$

$$L_t : \quad \frac{\beta^t \gamma}{1 - L_t^s} = \lambda_t w_t$$

$$K_{t+1} : \quad \lambda_t = \mathbb{E}_t \{ \lambda_{t+1} (r_{t+1} + 1 - \delta) \}$$

$$\lambda_t : \quad C_t + K_{t+1}^s - (1 - \delta)K_t = w_t L_t^s + r_t K_t - T_t$$

Combining the first-order conditions w.r.t. C_t and L_t we get:

$$\frac{\gamma C_t}{1 - L_t^s} = w_t$$

Combining the first-order conditions w.r.t. C_t and K_{t+1} we get the Euler Equation for consumption:

$$1 = \mathbb{E}_t \left\{ \beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

Equilibrium requires:

$$\left. \begin{aligned} K_t^d &= K_t^s = K_t \\ L_t^d &= L_t^s = L_t \end{aligned} \right\} \text{factor markets}$$

$$\left. \begin{aligned} Y_t &= C_t + I_t + G_t \\ Y_t &= \theta_t K_t^\alpha L_t^{1-\alpha} \\ I_t &= K_{t+1} - (1 - \delta)K_t \end{aligned} \right\} \text{output market}$$

$$\left. \begin{aligned} r_t &= \alpha \frac{Y_t}{K_t} \\ w_t &= (1 - \alpha) \frac{Y_t}{L_t} \end{aligned} \right\} \text{profit max}$$

$$\left. \begin{aligned} \frac{\gamma C_t}{1 - L_t} &= w_t \\ 1 &= \mathbb{E}_t \left\{ \beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\} \end{aligned} \right\} \text{utility max}$$

$$T_t = G_t \} \text{ gov. budget constraint}$$

Remark 1. Equilibrium in the market for output together with the zero-profit condition and the government's budget constraint ensure that the budget constraint of the household is satisfied.

Remark 2. We don't actually need to solve for T_t , since it's always equal to G_t .

Hence, we would like to solve the following system of nonlinear stochastic dynamic equations:

$$Y_t = \theta_t K_t^\alpha L_t^{1-\alpha}$$

$$w_t = (1 - \alpha) \frac{Y_t}{L_t}$$

$$r_t = \alpha \frac{Y_t}{K_t}$$

$$Y_t = C_t + I_t + G_t$$

$$I_t = K_{t+1} - (1 - \delta)K_t$$

$$\frac{\gamma C_t}{1 - L_t} = w_t$$

$$1 = \mathbb{E}_t \left\{ \beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

$$\ln \theta_{t+1} = (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1}$$

$$\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \eta_{t+1}$$

We have 9 equations per period to solve for the evolution of 9 variables: 7 endog. variables ($K_t, Y_t, C_t, I_t, L_t, r_t, w_t$) and 2 exogenous variables (θ_t, G_t). Of the 7 endogenous variables, one is a state variable (K_t) and six are jump variables ($Y_t, C_t, I_t, L_t, r_t, w_t$).

Nonstochastic Steady State

We use upper bars to denote nonstochastic steady-state values.

Setting $\varepsilon_{t+1} = 0$ and $\theta_{t+1} = \theta_t$ into $\ln \theta_{t+1} = (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1}$ we get

$$\bar{\theta} = \theta$$

Setting $\eta_{t+1} = 0$ and $G_{t+1} = G_t$ into $\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \eta_{t+1}$ we get

$$\bar{G} = G$$

Setting $C_t = C_{t+1}$ and $r_{t+1} = \bar{r}$ into the stochastic Euler equation we get:

$$\begin{aligned}\bar{r} &= \frac{1}{\beta} - 1 + \delta \\ &= \rho + \delta\end{aligned}$$

We know $r_t = \alpha \frac{Y_t}{K_t} = \alpha \theta_t \left(\frac{K_t}{L_t} \right)^{\alpha-1}$. Setting $r_t = \bar{r}$ and $\theta_t = \bar{\theta}$ we get

$$\overline{\left(\frac{K}{L} \right)} = \left(\frac{\alpha \bar{\theta}}{\bar{r}} \right)^{\frac{1}{1-\alpha}}$$

We know $w_t = (1 - \alpha) \frac{Y_t}{L_t} = (1 - \alpha) \theta_t \left(\frac{K_t}{L_t} \right)^{\alpha}$. Then:

$$\begin{aligned} \bar{w} &= (1 - \alpha) \bar{\theta} \overline{\left(\frac{K}{L} \right)}^{\alpha} \\ &= (1 - \alpha) \bar{\theta}^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \end{aligned}$$

We finish in two steps. First we express the rest of the variables in terms of $\bar{K}$ (and other known variables/parameters). Then we solve for $\bar{K}$.

Setting $K_{t+1} = K_t$ into $I_t = K_{t+1} - (1 - \delta)K_t$ we get:

$$\bar{I} = \delta \bar{K}$$

From $\overline{\left(\frac{K}{L}\right)} = \left(\frac{\alpha\bar{\theta}}{\bar{r}}\right)^{\frac{1}{1-\alpha}}$ we get:

$$\bar{L} = \left(\frac{\bar{r}}{\alpha\bar{\theta}}\right)^{\frac{1}{1-\alpha}} \bar{K}$$

From the production function: $\bar{Y} = \bar{\theta}\bar{K}^\alpha\bar{L}^{1-\alpha}$. Using the expression above: we get:

$$\bar{Y} = \frac{\bar{r}}{\alpha}\bar{K}$$

From $Y_t = C_t + I_t + G_t$ we get: $\bar{C} = \bar{Y} - \bar{I} - \bar{G}$.

Using the previous results: $\bar{C} = \frac{\bar{r}}{\alpha}\bar{K} - \delta\bar{K} - \bar{G}$.

Then:

$$\bar{C} = \left(\frac{\bar{r}}{\alpha} - \delta\right)\bar{K} - \bar{G}$$

From $\frac{\gamma C_t}{1-L_t} = w_t$ we get: $\gamma \bar{C} = \bar{w}(1 - \bar{L})$.

Substituting our previous results in the expression above:

$$\gamma \left[\left(\frac{\bar{r}}{\alpha} - \delta \right) \bar{K} - \bar{G} \right] = \bar{w} \left[1 - \left(\frac{\bar{r}}{\alpha \bar{\theta}} \right)^{\frac{1}{1-\alpha}} \bar{K} \right]$$

Solving for $\bar{K}$ we get

$$\bar{K} = \alpha \frac{\bar{w} + \gamma \bar{G}}{(\gamma + 1 - \alpha) \bar{r} - \gamma \alpha \delta}$$

where $\bar{G}$, $\bar{r}$ and $\bar{w}$ are already known.

Notice that the denominator is positive, so $\bar{K} > 0$.

Now we can substitute the value of $\bar{K}$ into our previous expressions to obtain $\bar{I}$, $\bar{L}$, $\bar{Y}$, and $\bar{C}$.

Solution Strategy

1. Log-linearize the equations characterizing the equilibrium of the system around the nonstochastic steady state. This gives a new system of linear stochastic difference equations whose solution approximates the one of the original nonlinear system (as long as we are close enough to the nonstochastic steady state).
2. Write the linear system in an appropriate matrix form.
3. Calibrate the model.
4. Solve for the recursive equilibrium law of motion using the method of undetermined coefficients.
5. Analyze the solution via impulse-response analysis and second-order properties (variances, covariances, etc.). If needed, detrend the variables using an appropriate filter (e.g., the Hodrick-Prescott filter).

Log-linearization

We follow Uhlig's approach to log-linearization (Uhlig, 1999). See the Appendix for a different approach.

For any variable X define the period- t log-deviation from its nonstochastic steady-state value:

$$\widetilde{X}_t \equiv \ln X_t - \ln \overline{X}$$

Then:

$$X_t = \overline{X} e^{\widetilde{X}_t}$$

We establish a few approximations that will be useful later on. Recall that a first-order Taylor approximation of $f(x, y)$ around $(x, y) = (0, 0)$ gives:

$$\begin{aligned} f(x, y) &\cong f(0, 0) + f_x(0, 0)(x - 0) + f_y(0, 0)(y - 0) \\ &= f(0, 0) + f_x(0, 0)x + f_y(0, 0)y \end{aligned}$$

Claim 1 :

$$e^{\widetilde{X}_t + a\widetilde{Y}_t} \cong 1 + \widetilde{X}_t + a\widetilde{Y}_t$$

Proof. A first-order Taylor approximation of $e^{\widetilde{X}_t + a\widetilde{Y}_t}$ around $(\widetilde{X}_t, \widetilde{Y}_t) = (0, 0)$ gives:

$$\begin{aligned} e^{\widetilde{X}_t + a\widetilde{Y}_t} &\cong e^{0+a0} + e^{0+a0}(\widetilde{X}_t - 0) + ae^{0+a0}(\widetilde{Y}_t - 0) \\ &= 1 + \widetilde{X}_t + a\widetilde{Y}_t \end{aligned}$$

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Corollary 1 :

$$e^{\widetilde{X}_t} \cong 1 + \widetilde{X}_t$$

Claim 2 :

$$\widetilde{X}_t \widetilde{Y}_t \cong 0$$

Proof. A first-order Taylor approximation of $\widetilde{X}_t \widetilde{Y}_t$ around $(\widetilde{X}_t, \widetilde{Y}_t) = (0, 0)$ gives:

$$\begin{aligned} \widetilde{X}_t \widetilde{Y}_t &\cong 0 + 0(\widetilde{X}_t - 0) + 0(\widetilde{Y}_t - 0) \\ &= 0 \end{aligned}$$

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Claim 3 :

$$\mathbb{E}_t\{ae^{\widetilde{X}_{t+1}}\} \cong a + a\mathbb{E}_t\{\widetilde{X}_{t+1}\}$$

Proof. Using $e^{\widetilde{X}_{t+1}} \cong 1 + \widetilde{X}_{t+1}$ inside the curly brackets:

$$\begin{aligned} \mathbb{E}_t\{ae^{\widetilde{X}_{t+1}}\} &\cong \mathbb{E}_t\{a(1 + \widetilde{X}_{t+1})\} \\ &= a + a\mathbb{E}_t\{\widetilde{X}_{t+1}\} \end{aligned}$$

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Now we log-linearize the 9 equations that characterize the equilibrium of our model.

For the technology shock we get:

$$\begin{aligned}\tilde{\theta}_{t+1} &\equiv \ln \theta_{t+1} - \ln \bar{\theta} \\ &= (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1} - \ln \bar{\theta} \\ &= (1 - \psi) \ln \bar{\theta} + \psi \ln \theta_t + \varepsilon_{t+1} - \ln \bar{\theta} \\ &= \psi (\ln \theta_t - \ln \bar{\theta}) + \varepsilon_{t+1} \\ &= \psi \tilde{\theta}_t + \varepsilon_{t+1}\end{aligned}$$

Analogously, for government expenditure we get:

$$\tilde{G}_{t+1} = \phi \tilde{G}_t + \eta_{t+1}$$

From the production function we get:

$$\begin{aligned}
\tilde{Y}_t &\equiv \ln Y_t - \ln \bar{Y} \\
&= \ln \left(\theta_t K_t^\alpha L_t^{1-\alpha} \right) - \ln \left(\bar{\theta} \bar{K}^\alpha \bar{L}^{1-\alpha} \right) \\
&= \ln \theta_t + \alpha \ln K_t + (1 - \alpha) \ln L_t \\
&\quad - \ln \bar{\theta} - \alpha \ln \bar{K} - (1 - \alpha) \ln \bar{L} \\
&= \left(\ln \theta_t - \ln \bar{\theta} \right) + \alpha \left(\ln K_t - \ln \bar{K} \right) \\
&\quad + (1 - \alpha) \left(\ln L_t - \ln \bar{L} \right) \\
&= \tilde{\theta}_t + \alpha \tilde{K}_t + (1 - \alpha) \tilde{L}_t
\end{aligned}$$

From $w_t = (1 - \alpha) \frac{Y_t}{L_t}$ we get:

$$\begin{aligned}
\tilde{w}_t &\equiv \ln w_t - \ln \bar{w} \\
&= \ln \left((1 - \alpha) \frac{Y_t}{L_t} \right) - \ln \left((1 - \alpha) \frac{\bar{Y}}{\bar{L}} \right) \\
&= \ln(1 - \alpha) + \ln Y_t - \ln L_t \\
&\quad - \ln(1 - \alpha) - \ln \bar{Y} + \ln \bar{L} \\
&= \left(\ln Y_t - \ln \bar{Y} \right) - \left(\ln L_t - \ln \bar{L} \right) \\
&= \tilde{Y}_t - \tilde{L}_t
\end{aligned}$$

Analogously, from $r_t = \alpha \frac{Y_t}{K_t}$ we get:

$$\tilde{r}_t = \tilde{Y}_t - \tilde{K}_t$$

Notice that the previous expressions are exact relations. No approximations were needed to derive them.

The following log-linearizations make use of the approximations derived previously (we keep using the $=$ sign instead of $\cong$ just for simplicity).

From $Y_t = C_t + I_t + G_t$ we get:

$$\overline{Y}e^{\tilde{Y}_t} = \overline{C}e^{\tilde{C}_t} + \overline{I}e^{\tilde{I}_t} + \overline{G}e^{\tilde{G}_t}$$

$$\overline{Y}(1 + \tilde{Y}_t) = \overline{C}(1 + \tilde{C}_t) + \overline{I}(1 + \tilde{I}_t) + \overline{G}(1 + \tilde{G}_t)$$

$$\overline{Y} + \overline{Y}\tilde{Y}_t = \overline{C} + \overline{C}\tilde{C}_t + \overline{I} + \overline{I}\tilde{I}_t + \overline{G} + \overline{G}\tilde{G}_t$$

$$\overline{Y} + \overline{Y}\tilde{Y}_t = (\overline{C} + \overline{I} + \overline{G}) + \overline{C}\tilde{C}_t + \overline{I}\tilde{I}_t + \overline{G}\tilde{G}_t$$

$$\overline{Y} + \overline{Y}\tilde{Y}_t = \overline{Y} + \overline{C}\tilde{C}_t + \overline{I}\tilde{I}_t + \overline{G}\tilde{G}_t$$

Then:

$$\overline{Y}\tilde{Y}_t = \overline{C}\tilde{C}_t + \overline{I}\tilde{I}_t + \overline{G}\tilde{G}_t$$

From $I_t = K_{t+1} - (1 - \delta)K_t$ we get:

$$\bar{I}e^{\tilde{I}_t} = \bar{K}e^{\tilde{K}_{t+1} - (1-\delta)\tilde{K}_t}$$

$$\bar{I}(1 + \tilde{I}_t) = \bar{K}(1 + \tilde{K}_{t+1}) - (1 - \delta)\bar{K}(1 + \tilde{K}_t)$$

$$\bar{I} + \bar{I}\tilde{I}_t = \bar{K} + \bar{K}\tilde{K}_{t+1} - (1 - \delta)\bar{K} - (1 - \delta)\bar{K}\tilde{K}_t$$

$$\bar{I} + \bar{I}\tilde{I}_t = \delta\bar{K} + \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

$$\bar{I} + \bar{I}\tilde{I}_t = \bar{I} + \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

$$\bar{I}\tilde{I}_t = \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

$$\delta\bar{K}\tilde{I}_t = \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

Then:

$$\delta\tilde{I}_t = \tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t$$

From $\frac{\gamma C_t}{1-\bar{L}_t} = w_t$ we get:

$$\gamma C_t = w_t(1 - \bar{L}_t)$$

$$\gamma \bar{C} e^{\tilde{C}_t} = \bar{w} e^{\tilde{w}_t} (1 - \bar{L} e^{\tilde{L}_t})$$

$$\gamma \bar{C} (1 + \tilde{C}_t) = \bar{w} (1 + \tilde{w}_t) [1 - \bar{L} (1 + \tilde{L}_t)]$$

$$\gamma \bar{C} + \gamma \bar{C} \tilde{C}_t = (\bar{w} + \bar{w} \tilde{w}_t) [(1 - \bar{L}) - \bar{L} \tilde{L}_t]$$

$$\gamma \bar{C} + \gamma \bar{C} \tilde{C}_t = \bar{w} (1 - \bar{L}) - \bar{w} \bar{L} \tilde{L}_t + \bar{w} \tilde{w}_t (1 - \bar{L}) - \bar{w} \tilde{w}_t \bar{L} \tilde{L}_t$$

$$\bar{w} (1 - \bar{L}) + \bar{w} (1 - \bar{L}) \tilde{C}_t = \bar{w} (1 - \bar{L}) - \bar{w} \bar{L} \tilde{L}_t + \bar{w} (1 - \bar{L}) \tilde{w}_t - \bar{w} \bar{L} \tilde{w}_t \tilde{L}_t$$

$$\bar{w} (1 - \bar{L}) \tilde{C}_t = -\bar{w} \bar{L} \tilde{L}_t + \bar{w} (1 - \bar{L}) \tilde{w}_t$$

$$\tilde{C}_t = -\frac{\bar{w} \bar{L}}{\bar{w} (1 - \bar{L})} \tilde{L}_t + \frac{\bar{w} (1 - \bar{L})}{\bar{w} (1 - \bar{L})} \tilde{w}_t$$

Then:

$$\tilde{C}_t = \tilde{w}_t - \frac{\bar{L}}{1 - \bar{L}} \tilde{L}_t$$

From $1 = \mathbb{E}_t\{\beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta)\}$ we get:

$$1 = \mathbb{E}_t \left\{ \beta \frac{\bar{C}_e \tilde{C}_t}{\bar{C}_e \tilde{C}_{t+1}} \left(\bar{r} e^{\tilde{r}_{t+1}} + 1 - \delta \right) \right\}$$

$$1 = \mathbb{E}_t \{ \beta e^{\tilde{C}_t - \tilde{C}_{t+1}} [\bar{r}(1 + \tilde{r}_{t+1}) + 1 - \delta] \}$$

$$1 = \mathbb{E}_t \{ \beta (1 + \tilde{C}_t - \tilde{C}_{t+1}) [\bar{r} + \bar{r} \tilde{r}_{t+1} + 1 - \delta] \}$$

$$1 = \mathbb{E}_t \{ \beta (1 + \tilde{C}_t - \tilde{C}_{t+1}) [(\bar{r} + 1 - \delta) + \bar{r} \tilde{r}_{t+1}] \}$$

$$1 = \mathbb{E}_t \{ (\beta + \beta \tilde{C}_t - \beta \tilde{C}_{t+1}) (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) \}$$

$$1 = \mathbb{E}_t \{ \beta (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) + \beta \tilde{C}_t (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) - \beta \tilde{C}_{t+1} (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) \}$$

$$1 = \mathbb{E}_t \{ 1 + \beta \bar{r} \tilde{r}_{t+1} + \tilde{C}_t + \beta \bar{r} \tilde{C}_t \tilde{r}_{t+1} - \tilde{C}_{t+1} - \beta \bar{r} \tilde{C}_{t+1} \tilde{r}_{t+1} \}$$

$$1 = \mathbb{E}_t \{ 1 + \beta \bar{r} \tilde{r}_{t+1} + \tilde{C}_t - \tilde{C}_{t+1} \}$$

$$0 = \mathbb{E}_t \{ \beta \bar{r} \tilde{r}_{t+1} + \tilde{C}_t - \tilde{C}_{t+1} \}$$

Then:

$$0 = \mathbb{E}_t \{ \tilde{C}_{t+1} - \beta \bar{r} \tilde{r}_{t+1} - \tilde{C}_t \}$$

Summarizing

$$\tilde{Y}_t = \tilde{\theta}_t + \alpha \tilde{K}_t + (1 - \alpha) \tilde{L}_t$$

$$\tilde{w}_t = \tilde{Y}_t - \tilde{L}_t$$

$$\tilde{r}_t = \tilde{Y}_t - \tilde{K}_t$$

$$\bar{Y} \tilde{Y}_t = \bar{C} \tilde{C}_t + \bar{I} \tilde{I}_t + \bar{G} \tilde{G}_t$$

$$\delta \tilde{I}_t = \tilde{K}_{t+1} - (1 - \delta) \tilde{K}_t$$

$$\tilde{C}_t = \tilde{w}_t - \frac{\bar{L}}{1 - \bar{L}} \tilde{L}_t$$

$$0 = \mathbb{E}_t \left\{ \tilde{C}_{t+1} - \beta \bar{r} \tilde{r}_{t+1} - \tilde{C}_t \right\}$$

$$\tilde{\theta}_{t+1} = \psi \tilde{\theta}_t + \varepsilon_{t+1}$$

$$\tilde{G}_{t+1} = \phi \tilde{G}_t + \eta_{t+1}$$

where $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and $\eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$ (independent from each other).

Matrix Form

Define the following column vectors:

$$x_t \equiv [\widetilde{K}_{t+1}] , \quad y_t \equiv \begin{bmatrix} \widetilde{Y}_t \\ \widetilde{C}_t \\ \widetilde{I}_t \\ \widetilde{L}_t \\ \widetilde{r}_t \\ \widetilde{w}_t \end{bmatrix} , \quad z_t \equiv \begin{bmatrix} \widetilde{\theta}_t \\ \widetilde{G}_t \end{bmatrix} , \quad \epsilon_{t+1} \equiv \begin{bmatrix} \varepsilon_{t+1} \\ \eta_{t+1} \end{bmatrix}$$

Then, we can rewrite our log-linearized system in matrix form, as follows:

$$\mathbf{0} = Ax_t + Bx_{t-1} + Cy_t + Dz_t$$

$$\mathbf{0} = \mathbb{E}_t \left\{ \begin{array}{l} Fx_{t+1} + Gx_t + Hx_{t-1} + \\ Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t \end{array} \right\}$$

$$z_{t+1} = Nz_t + \epsilon_{t+1} \quad \text{with} \quad \mathbb{E}_t\{\epsilon_{t+1}\} = \mathbf{0}$$

where

$$A = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad B = \begin{bmatrix} -\alpha \\ 0 \\ -1 \\ 0 \\ -(1-\delta) \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & -(1-\alpha) & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & 0 \\ \bar{Y} & -\bar{C} & -\bar{I} & 0 & 0 & 0 \\ 0 & 0 & -\delta & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{\bar{L}}{1-\bar{L}} & 0 & -1 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -\overline{G} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$F = [0], \quad G = [0], \quad H = [0]$$

$$J = \begin{bmatrix} 0 & 1 & 0 & 0 & -\beta \bar{r} & 0 \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} 0 & 0 \end{bmatrix}, \quad M = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$N = \begin{bmatrix} \psi & 0 \\ 0 & \phi \end{bmatrix}$$

Calibration

For quarterly data (similar to Hansen & Wright, 1992):

$$\alpha = 0.36$$

$$\beta = 0.99$$

$$\gamma = 2.5$$

$$\delta = 0.025$$

$$\psi = 0.95$$

$$\phi = 0.96$$

$$\bar{\theta} = 1$$

$$\bar{G} = 0.27$$

$$\sigma_{\varepsilon} = 0.007$$

$$\sigma_{\eta} = 0.021$$

Then:

$$\bar{r} = 0.035$$

$$\bar{w} = 2.37$$

$$\overline{K} = 12.5$$

$$\overline{L} = 0.329$$

$$\overline{Y} = 1.219$$

$$\overline{I} = 0.312$$

$$\overline{C} = 0.636$$

$$\frac{\overline{G}}{\overline{\overline{Y}}} = 0.22$$

$$\frac{\overline{I}}{\overline{\overline{Y}}} = 0.26$$

$$\frac{\overline{K}}{4\overline{\overline{Y}}} = 2.56$$

Solution (Uhlig, 1999)

We look for recursive law of motions

$$x_t = Px_{t-1} + Qz_t$$

$$y_t = Rx_{t-1} + Sz_t$$

with the property that the equilibrium described by these rules is *stable*.

Solution strategy: solve for P , Q , R , and S using the method of undetermined coefficients.

We assume that the matrix C in the block of nonexpectational equations is invertible. For the more general case see Uhlig (1999).

We start by substituting $y_t = Rx_{t-1} + Sz_t$ into the first block $0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t$. Then:

$$0 = Ax_t + Bx_{t-1} + C(Rx_{t-1} + Sz_t) + Dz_t$$

$$0 = Ax_t + Bx_{t-1} + CRx_{t-1} + CSz_t + Dz_t$$

$$0 = Ax_t + (B + CR)x_{t-1} + (D + CS)z_t$$

Then:

$$Ax_t = -(B + CR)x_{t-1} - (D + CS)z_t$$

From $x_t = Px_{t-1} + Qz_t$ we get:

$$Ax_t = APx_{t-1} + AQz_t$$

For the right-hand sides of the two expressions above to coincide *for every pair* (x_{t-1}, z_t) , we need:

$$AP = -(B + CR)$$

$$AQ = -(D + CS)$$

From the first expression we get

$$R = -C^{-1}(AP + B)$$

and from the second

$$S = -C^{-1}(AQ + D)$$

Hence, we've managed to express R in terms of P , and S in terms of Q . Now we have to find P and Q .

Substituting $x_t = Px_{t-1} + Qz_t$, $y_t = Rx_{t-1} + Sz_t$, and $z_{t+1} = Nz_t + \epsilon_{t+1}$ into $\mathbf{0} = \mathbb{E}_t\{Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t\}$ we get:

$$\mathbf{0} = \mathbb{E}_t \left\{ \begin{array}{l} F(Px_t + Qz_{t+1}) + G(Px_{t-1} + Qz_t) + Hx_{t-1} \\ + J(Rx_t + Sz_{t+1}) + Ky_t + Lz_{t+1} + Mz_t \end{array} \right\}$$

$$\mathbf{0} = \mathbb{E}_t \left\{ \begin{array}{l} FPx_t + FQz_{t+1} + GPx_{t-1} + GQz_t + Hx_{t-1} \\ + JRx_t + JSz_{t+1} + Ky_t + Lz_{t+1} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQ\mathbb{E}_t\{z_{t+1}\} + GPx_{t-1} + GQz_t + Hx_{t-1} + \\ JRx_t + JS\mathbb{E}_t\{z_{t+1}\} + Ky_t + L\mathbb{E}_t\{z_{t+1}\} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQ\mathbb{E}_t\{Nz_t + \epsilon_{t+1}\} + GPx_{t-1} + \\ GQz_t + Hx_{t-1} + JRx_t + JS\mathbb{E}_t\{Nz_t + \epsilon_{t+1}\} + \\ Ky_t + L\mathbb{E}_t\{Nz_t + \epsilon_{t+1}\} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQNz_t + FQ\mathbb{E}_t\{\epsilon_{t+1}\} + GPx_{t-1} + \\ GQz_t + Hx_{t-1} + JRx_t + JSNz_t + JS\mathbb{E}_t\{\epsilon_{t+1}\} + \\ Ky_t + LNz_t + L\mathbb{E}_t\{\epsilon_{t+1}\} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQNz_t + GPx_{t-1} + \\ GQz_t + Hx_{t-1} + JRx_t + JSNz_t + \\ Ky_t + LNz_t + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP + JR)x_t + \\ (FQN + GQ + JSN + M + LN)z_t + \\ (GP + H)x_{t-1} + Ky_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP + JR)(Px_{t-1} + Qz_t) + \\ (FQN + GQ + JSN + M + LN)z_t + \\ (GP + H)x_{t-1} + Ky_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP + JR)Px_{t-1} + (FP + JR)Qz_t + \\ (FQN + GQ + JSN + M + LN)z_t + \\ (GP + H)x_{t-1} + Ky_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP^2 + JRP + GP + H)x_{t-1} + \\ [(FP + JR)Q + FQN + GQ + JSN + M + LN]z_t + \\ Ky_t \end{array} \right\}$$

Then:

$$\begin{aligned} Ky_t = & -[FP^2 + (JR + G)P + H]x_{t-1} \\ & -[(FP + JR + G)Q + FQN + JSN + LN + M]z_t \end{aligned}$$

From $y_t = Rx_{t-1} + Sz_t$ we get:

$$Ky_t = KRx_{t-1} + KSz_t$$

Comparing the previous two expressions we conclude:

$$KR = -[FP^2 + (JR + G)P + H]$$

$$KS = -[(FP + JR + G)Q + FQN + JSN + LN + M]$$

From the first of the two expressions above we obtain:

$KR = -FP^2 - JRP - GP - H$. Then we can use $R = -C^{-1}(AP + B)$ to eliminate R . We get:

$$-KC^{-1}(AP+B) = -FP^2 + JC^{-1}(AP+B)P - GP - H$$

$$-KC^{-1}AP - KC^{-1}B = -FP^2 + JC^{-1}AP^2 + JC^{-1}BP - GP - H$$

$$0 = -(F - JC^{-1}A)P^2 + (JC^{-1}B - G + KC^{-1}A)P - H + KC^{-1}B$$

Then:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P + (H - KC^{-1}B) = 0$$

The previous expression is a matrix quadratic equation in P . Once the model is calibrated, this equation can be solved with a computer. There are different solution methods available (see the Appendix).

Stability requires that every eigenvalue of P has modulus smaller than one. See the Appendix and Uhlig (1999) for a discussion.

Notice that, once we have found P , we can obtain R from $R = -C^{-1}(AP + B)$.

Now we need to solve for Q using the fact that

$$KS = -[(FP + JR + G)Q + FQN + JSN + LN + M]$$

Notice that we have already found P and R .

We start by rewriting the expression above as follows:

$$KS = -(FP + JR + G)Q - FQN - JSN - LN - M$$

Using $S = -C^{-1}(AQ + D)$ to eliminate S we get:

$$-KC^{-1}(AQ + D) = -(FP + JR + G)Q - FQN + JC^{-1}(AQ + D)N - LN - M$$

$$-KC^{-1}AQ - KC^{-1}D = -(FP + JR + G)Q - FQN + JC^{-1}AQN + JC^{-1}DN - LN - M$$

$$[FP + JR + G - KC^{-1}A]Q - KC^{-1}D = -(F - JC^{-1}A)QN + JC^{-1}DN - LN - M$$

Then:

$$(F - JC^{-1}A)QN + [FP + JR + G - KC^{-1}A]Q = JC^{-1}DN - LN - M + KC^{-1}D$$

Notice that the different ways in which Q enters in the left-hand side of the expression above prevents us from solving for Q directly. We can, however, solve for Q using the **vec operator** (column-wise vectorization).

Aside on the vec operator

Let A be the 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Then:

$$\text{vec}(A) = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{bmatrix}$$

The previous definition generalizes in the obvious way to any matrix.

Theorem 1. $\text{vec}(A + B) = \text{vec}(A) + \text{vec}(B)$

Theorem 2. $\text{vec}(ABC) = (C' \otimes A)\text{vec}(B)$

Corollary: Setting $C = I$ (identity matrix) in Theorem 2 we get: $\text{vec}(AB) = (I \otimes A)\text{vec}(B)$

Reminder. The symbol $\otimes$ denotes de Kronecker product between two matrices.

For example, if

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

and

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

we get:

$$\begin{aligned} A \otimes B &= \begin{bmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{bmatrix} \\ &= \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{bmatrix} \end{aligned}$$

Recall we have:

$$\begin{aligned} & (F - JC^{-1}A)QN + [FP + JR + G - KC^{-1}A]Q \\ &= JC^{-1}DN - LN - M + KC^{-1}D \end{aligned}$$

Applying the *vec* operator to both sides of the expression above we get:

$$\begin{aligned} & \text{vec}\{(F - JC^{-1}A)QN + [FP + JR + G - KC^{-1}A]Q\} \\ &= \text{vec}(JC^{-1}DN - LN - M + KC^{-1}D) \end{aligned}$$

$$\begin{aligned} & \text{vec}\{(F - JC^{-1}A)QN\} + \text{vec}\{[FP + JR + G - KC^{-1}A]Q\} \\ &= \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \quad \text{by Th. 1} \end{aligned}$$

$$\begin{aligned} & [N' \otimes (F - JC^{-1}A)]\text{vec}(Q) + [I \otimes (FP + JR + G - KC^{-1}A)]\text{vec}(Q) \\ &= \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \quad \text{by Th.2 and its Corollary} \end{aligned}$$

$$\begin{aligned} & [N' \otimes (F - JC^{-1}A) + I \otimes (FP + JR + G - KC^{-1}A)]\text{vec}(Q) \\ &= \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \end{aligned}$$

Then:

$$\begin{aligned} \text{vec}(Q) = & [N' \otimes (F - JC^{-1}A) + I \otimes (FP + JR + G - KC^{-1}A)]^{-1} \\ & \times \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \end{aligned}$$

provided the inverse exists.

From $\text{vec}(Q)$ we can recover Q .

Having found Q we can obtain S from

$$S = -C^{-1}(AQ + D)$$

Hence, we've managed to solve for P , Q , R , and S .

Now we are ready to solve our calibrated model using Uhlig's Matlab code (usually called Uhlig's Toolkit).

Notational convention. To use Uhlig's toolkit we have to change the time convention for the capital stock. What we have called K_{t+1} ($\widetilde{K}_{t+1}$) has to be renamed as K_t ($\widetilde{K}_t$). Analogously, K_t ($\widetilde{K}_t$) has to be changed to K_{t-1} ($\widetilde{K}_{t-1}$).

After running the code we get:

$$P = [0.9561]$$

$$Q = [0.1051 \quad -0.0032]$$

$$R = \begin{bmatrix} 0.1437 \\ 0.6474 \\ -0.7577 \\ -0.3380 \\ -0.8563 \\ 0.4817 \end{bmatrix}, \quad S = \begin{bmatrix} 1.3543 & 0.1111 \\ 0.5293 & -0.1477 \\ 4.2041 & -0.1299 \\ 0.5535 & 0.1737 \\ 1.3543 & 0.1111 \\ 0.8007 & -0.0625 \end{bmatrix}$$

Then, $x_t = Px_{t-1} + Qz_t$ becomes:

$$\widetilde{K}_{t+1} = 0.9561\widetilde{K}_t + 0.1051\widetilde{\theta}_t - 0.0032\widetilde{G}_t$$

Also, $y_t = Rx_{t-1} + Sz_t$ gives:

$$\widetilde{Y}_t = 0.1437\widetilde{K}_t + 1.3543\widetilde{\theta}_t + 0.1111\widetilde{G}_t$$

$$\widetilde{C}_t = 0.6474\widetilde{K}_t + 0.5293\widetilde{\theta}_t - 0.1477\widetilde{G}_t$$

$$\widetilde{I}_t = -0.7577\widetilde{K}_t + 4.2041\widetilde{\theta}_t - 0.1299\widetilde{G}_t$$

$$\widetilde{L}_t = -0.3380\widetilde{K}_t + 0.5535\widetilde{\theta}_t + 0.1737\widetilde{G}_t$$

$$\widetilde{r}_t = -0.8563\widetilde{K}_t + 1.3543\widetilde{\theta}_t + 0.1111\widetilde{G}_t$$

$$\widetilde{w}_t = 0.4817\widetilde{K}_t + 0.8007\widetilde{\theta}_t - 0.0625\widetilde{G}_t$$

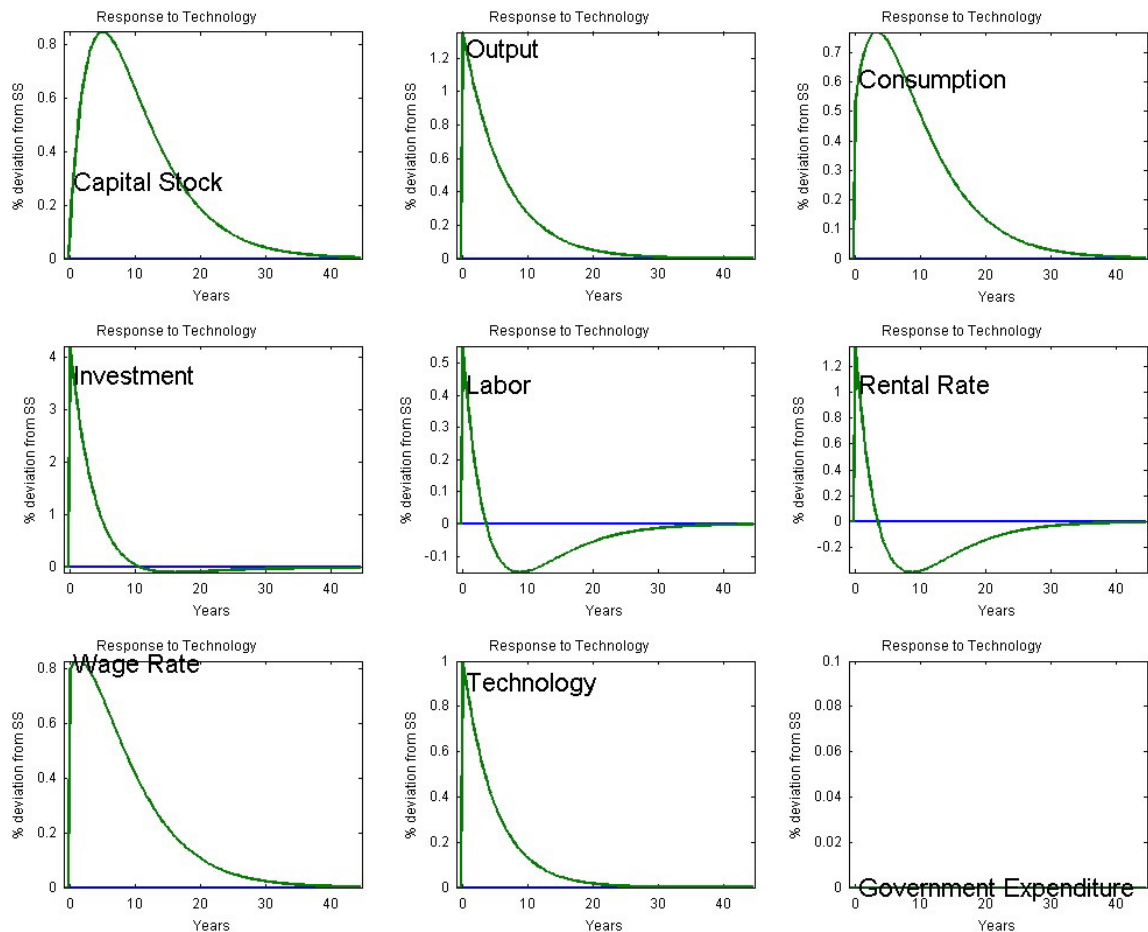
Finally, recall $z_{t+1} = Nz_t + \epsilon_{t+1}$ gives:

$$\widetilde{\theta}_{t+1} = 0.95\widetilde{\theta}_t + \varepsilon_{t+1} \quad \text{with } \varepsilon_{t+1} \sim \mathcal{N}(0, 0.007^2)$$

$$\widetilde{G}_{t+1} = 0.96\widetilde{G}_t + \eta_{t+1} \quad \text{with } \eta_{t+1} \sim \mathcal{N}(0, 0.021^2)$$

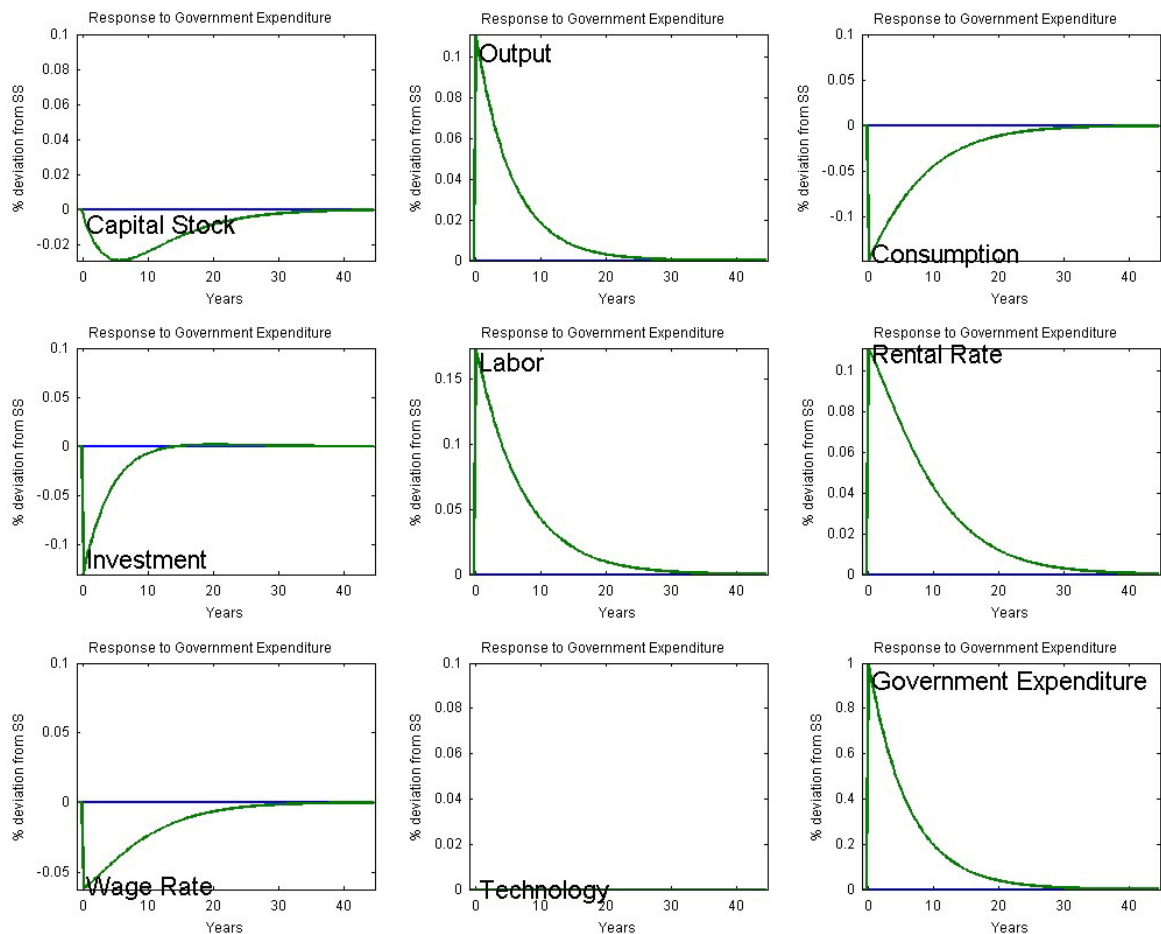
Impulse Responses

Response to a 1% Shock to Technology

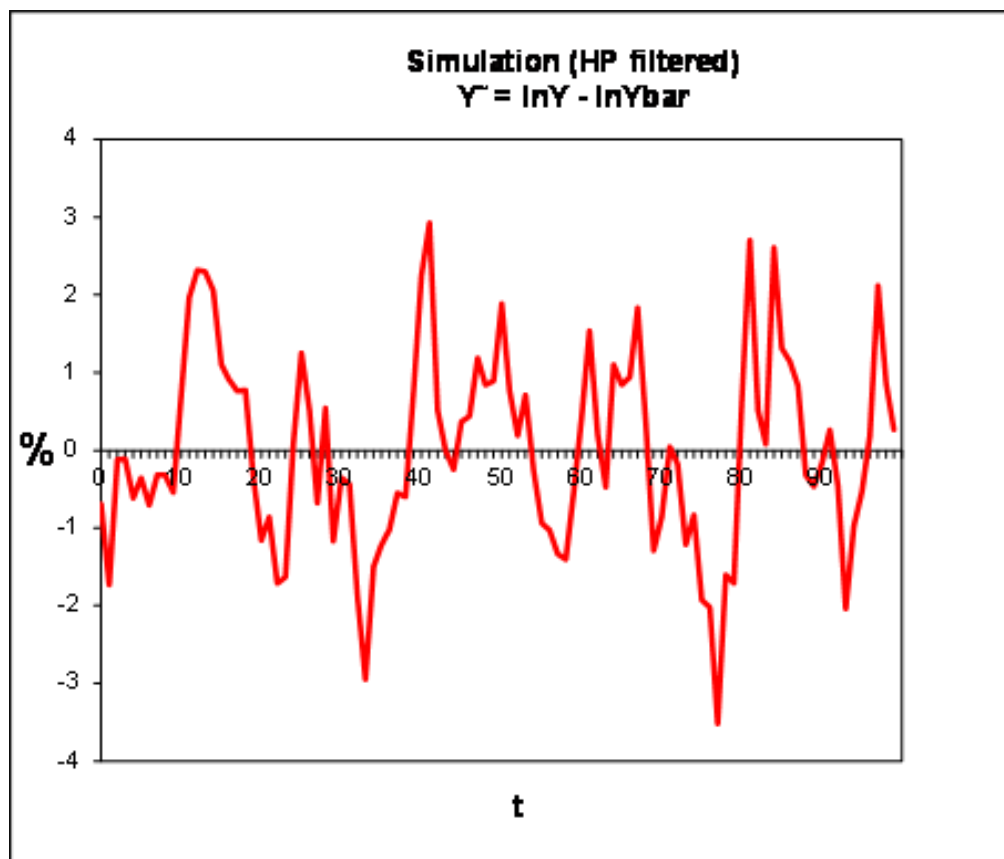


Impulse Responses

Response to a 1% Shock to Government Expenditure



Simulation for the cyclical component of output



Cyclical Properties of Time Series

Standard Deviations (%)

		Data	Model
Output	Y	1.92	1.28
Consumption	C	0.86	0.66
Investment	I	5.33	3.88
Labor (hours)	L	1.50	0.69
Wage	w	1.10	0.77

Note. The column "Data" reports figures for the U.S. over the period 1947:1-1991:3 (Hansen & Wright 1992, Table 2). The column "Model" reports the sample means of statistics computed for each of 100 simulations of our calibrated model. The length of each simulation is 179 periods (quarters), which coincides with the length of the sample in Hansen & Wright (1992). Similar figures are reported by Hansen & Wright (1992) in Table 3. Both, U.S. data and model data were detrended using the HP filter with $\lambda = 1600$.

Standard Deviations Relative to Output

		Data	Model
Output	Y	1.00	1.00
Consumption	C	0.45	0.52
Investment	I	2.78	3.04
Labor (hours)	L	0.78	0.54
Wage	w	0.57	0.61

Contemporaneous Correlations with Output

		Data	Model
Output	Y	1.00	1.00
Consumption	C	0.71	0.59
Investment	I	0.73	0.94
Labor (hours)	L	0.82	0.85
Wage	w	0.63	0.88

Hours vs. Productivity

	Data	Model
σ_L/σ_w	1.37	0.90
$Corr(L, w)$	0.07	0.50

Recall that w is proportional to $\frac{Y}{L}$ (average productivity of labor). Hence $\tilde{w} = \widetilde{\left(\frac{Y}{L}\right)} = \tilde{Y} - \tilde{L}$.

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Appendix

Log-linearization

Consider the expression $y_t = f(x_t)$, where f is a differentiable function, and let $\bar{y} = f(\bar{x})$ be its steady-state value. We want to approximate the log-deviation $\ln y_t - \ln \bar{y}$ with a linear function of the log-deviation $\ln x_t - \ln \bar{x}$, for values of x_t close to $\bar{x}$.

From $y_t = f(x_t)$ we get: $\ln y_t = \ln f(x_t)$. We can rewrite this as follows: $\ln y_t = \ln f(e^{\ln x_t}) \equiv g(\ln x_t)$. Now we take a first-order Taylor approximation of $g(\ln x_t)$ around $\ln \bar{x}$:¹

$$g(\ln x_t) \cong g(\ln \bar{x}) + g'(\ln \bar{x})(\ln x_t - \ln \bar{x})$$

Using $g(\ln x_t) \equiv \ln f(e^{\ln x_t})$ we get:

$$\ln f(e^{\ln x_t}) \cong \ln f(e^{\ln \bar{x}}) + \frac{1}{f(e^{\ln \bar{x}})} f'(e^{\ln \bar{x}}) e^{\ln \bar{x}} (\ln x_t - \ln \bar{x})$$

We can rewrite the expression above as follows:

$$\begin{aligned} \ln f(x_t) &\cong \ln f(\bar{x}) + \frac{\bar{x}}{f(\bar{x})} f'(\bar{x}) (\ln x_t - \ln \bar{x}) \\ &= \ln f(\bar{x}) + e_f(\bar{x}) (\ln x_t - \ln \bar{x}) \end{aligned}$$

where $e_f(\bar{x}) \equiv \frac{\bar{x}}{f(\bar{x})} f'(\bar{x})$ is the elasticity of the function f evaluated at $\bar{x}$. Recalling that $y_t = f(x_t)$ we get the expression we were looking for:

$$\ln y_t - \ln \bar{y} \cong e_f(\bar{x}) (\ln x_t - \ln \bar{x})$$

Defining $\hat{y}_t \equiv \ln y_t - \ln \bar{y}$ and $\hat{x}_t \equiv \ln x_t - \ln \bar{x}$ we get:

$$\hat{y}_t \cong e_f(\bar{x}) \hat{x}_t$$

Notice that we can write: $\hat{x}_t \equiv \ln x_t - \ln \bar{x} = \ln \left(\frac{x_t}{\bar{x}} \right) = \ln \left(1 + \frac{x_t - \bar{x}}{\bar{x}} \right)$. For values of x_t sufficiently close to $\bar{x}$ we know that $\ln \left(1 + \frac{x_t - \bar{x}}{\bar{x}} \right) \cong \frac{x_t - \bar{x}}{\bar{x}}$ (this follows from a first-order Taylor approximation of $m(x_t) \equiv \ln \left(1 + \frac{x_t - \bar{x}}{\bar{x}} \right)$ around $\bar{x}$). Then: $\hat{x}_t \cong \frac{x_t - \bar{x}}{\bar{x}}$. Therefore, we can write:

$$\frac{y_t - \bar{y}}{\bar{y}} \cong e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}}$$

¹ An alternative approach to log-linearization is presented in: Uhlig, Harald (1999). "A Toolkit for Analysing Nonlinear Dynamic Stochastic Models Easily," in Ramon Marimon and Andrew Scott (eds.), *Computational Methods for the Study of Dynamic Economies*, Oxford: Oxford University Press, pp. 30-61.

In fact, we can derive the previous expression directly. First, write $y_t = f(x_t) = e^{\ln f(x_t)}$. Then: $\frac{y_t}{\bar{y}} = \frac{e^{\ln f(x_t)}}{e^{\ln f(\bar{x})}} = e^{\ln f(x_t) - \ln f(\bar{x})} \equiv h(x_t)$. A first-order Taylor approximation of $h(x_t)$ around $\bar{x}$ gives:

$$h(x_t) \cong h(\bar{x}) + h'(\bar{x})(x_t - \bar{x})$$

Using $h(x_t) \equiv e^{\ln f(x_t) - \ln f(\bar{x})}$ we get:

$$\begin{aligned} e^{\ln f(x_t) - \ln f(\bar{x})} &\cong e^0 + e^0 \frac{1}{f(\bar{x})} f'(\bar{x})(x_t - \bar{x}) \\ &= 1 + \frac{\bar{x}}{f(\bar{x})} f'(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \\ &= 1 + e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \end{aligned}$$

Finally, using $e^{\ln f(x_t) - \ln f(\bar{x})} = \frac{y_t}{\bar{y}}$ we obtain: $\frac{y_t}{\bar{y}} \cong 1 + e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \Rightarrow \frac{y_t}{\bar{y}} - 1 \cong e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \Rightarrow$

$$\frac{y_t - \bar{y}}{\bar{y}} \cong e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}}$$

which is what we wanted to show.

The previous expression suggests a simple way of log-linearizing $y_t = f(x_t)$ through differentiation. Start by taking the natural logarithm: $\ln y_t = \ln f(x_t)$. Then totally differentiate, evaluating all derivatives at their steady-state levels: $\frac{dy_t}{y} = \frac{f'(\bar{x})}{f(\bar{x})} dx_t \Rightarrow \frac{dy_t}{y} = \frac{\bar{x}}{f(\bar{x})} f'(\bar{x}) \frac{dx_t}{\bar{x}} \Rightarrow \frac{dy_t}{y} = e_f(\bar{x}) \frac{dx_t}{\bar{x}}$. Finally, interpret dx_t as $x_t - \bar{x}$ and dy_t as $y_t - \bar{y}$. Similarly, we can directly differentiate $y_t = f(x_t)$ to get $dy_t = f'(\bar{x}) dx_t$, and then divide by $\bar{y} = f(\bar{x})$ to obtain: $\frac{dy_t}{y} = \frac{f'(\bar{x})}{f(\bar{x})} dx_t$. Multiplying and dividing the right-hand side by $\bar{x}$ we get: $\frac{dy_t}{y} = \frac{\bar{x} f'(\bar{x})}{f(\bar{x})} \frac{dx_t}{\bar{x}} \Rightarrow \frac{dy_t}{y} = e_f(\bar{x}) \frac{dx_t}{\bar{x}}$.

A similar procedure works for multivariate functions, $y_t = f(x_{1t}, x_{2t}, \dots, x_{nt})$. In this case we get:

$$\hat{y}_t \cong e_{f,1}(\bar{x}) \hat{x}_{1t} + e_{f,2}(\bar{x}) \hat{x}_{2t} + \dots + e_{f,n}(\bar{x}) \hat{x}_{nt}$$

where $e_{f,i}(\bar{x})$ is the partial elasticity of f with respect to its i -th argument, evaluated at $\bar{x}$.

In stochastic models, we usually have expressions of the form

$$y_t = \mathbb{E}_t\{f(x_{t+1})\}$$

Taking natural logarithms we get: $\ln y_t = \ln \mathbb{E}_t\{f(x_{t+1})\}$. Now we make the approximation $\ln \mathbb{E}_t\{f(x_{t+1})\} \cong \mathbb{E}_t\{\ln f(x_{t+1})\}$ to get:²

$$\ln y_t \cong \mathbb{E}_t\{\ln f(x_{t+1})\}$$

²Since $\ln(\cdot)$ is a strictly concave function, we know from Jensen's Inequality that $\ln \mathbb{E}_t\{f(x_{t+1})\} > \mathbb{E}_t\{\ln f(x_{t+1})\}$.

Since $\ln \bar{y} = \ln f(\bar{x})$, the expression above gives: $\ln y_t - \ln \bar{y} \cong \mathbb{E}_t\{\ln f(x_{t+1})\} - \ln f(\bar{x}) \Rightarrow$

$$\ln y_t - \ln \bar{y} \cong \mathbb{E}_t\{\ln f(x_{t+1}) - \ln f(\bar{x})\}$$

Using $\ln f(x_{t+1}) \cong \ln f(\bar{x}) + e_f(\bar{x})(\ln x_{t+1} - \ln \bar{x})$ we can rewrite the expression above as follows: $\ln y_t - \ln \bar{y} \cong \mathbb{E}_t\{e_f(\bar{x})(\ln x_{t+1} - \ln \bar{x})\} \Rightarrow$

$$\hat{y}_t \cong e_f(\bar{x})\mathbb{E}_t\{\hat{x}_{t+1}\}$$

Log-linearizing a standard RBC model

The following equations describe a standard RBC model:³

$\ln \lambda_{t+1} = (1 - \gamma) \ln \lambda + \gamma \ln \lambda_t + \epsilon_{t+1}$	Evolution of TFP
$K_{t+1} = I_t + (1 - \delta)K_t$	Law of motion for capital
$Y_t = \lambda_t K_t^\theta H_t^{1-\theta}$	Cobb-Douglas technology
$w_t = (1 - \theta) \frac{Y_t}{H_t}$	FOC for labor (firm)
$r_t = \theta \frac{Y_t}{K_t}$	FOC for capital (firm)
$Y_t = C_t + I_t$	Resource constraint
$\frac{aC_t}{1-H_t} = w_t$	Intratemporal optimality condition (household)
$\frac{1}{C_t} = \mathbb{E}_t \left\{ \frac{\beta}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$	Euler Equation (household)

From $\ln \lambda_{t+1} = (1 - \gamma) \ln \lambda + \gamma \ln \lambda_t + \epsilon_{t+1}$, and $\bar{\lambda} = \lambda$ we get:

$$\begin{aligned} \hat{\lambda}_{t+1} &\equiv \ln \lambda_{t+1} - \ln \bar{\lambda} \\ &= (1 - \gamma) \ln \lambda + \gamma \ln \lambda_t + \epsilon_{t+1} - \ln \lambda \\ &= \gamma (\ln \lambda_t - \ln \lambda) + \epsilon_{t+1} \\ &= \gamma \hat{\lambda}_t + \epsilon_{t+1} \end{aligned}$$

Notice that this case does not require any approximation. The same is true for the production function and the two first-order conditions for the firm: $Y_t = \lambda_t K_t^\theta H_t^{1-\theta}$, $w_t = (1 - \theta) \frac{Y_t}{H_t}$, and $r_t = \theta \frac{Y_t}{K_t}$. Taking natural logs and subtracting the log of the steady-state values we get:

$$\begin{aligned} \hat{Y}_t &= \hat{\lambda}_t + \theta \hat{K}_t + (1 - \theta) \hat{H}_t \\ \hat{w}_t &= \hat{Y}_t - \hat{H}_t \\ \hat{r}_t &= \hat{Y}_t - \hat{K}_t \end{aligned}$$

³The per-period utility function is: $u(C_t, H_t) = \ln C_t + a \ln(1 - H_t)$.

From $K_{t+1} = I_t + (1 - \delta)K_t$ we get $dK_{t+1} = dI_t + (1 - \delta)dK_t$. Dividing by $\bar{K} = \frac{\bar{I}}{\delta}$ we get: $\frac{dK_{t+1}}{\bar{K}} = \delta \frac{dI_t}{\bar{I}} + (1 - \delta) \frac{dK_t}{\bar{K}} \Rightarrow$

$$\hat{K}_{t+1} = \delta \hat{I}_t + (1 - \delta) \hat{K}_t$$

From $Y_t = C_t + I_t$ we get $dY_t = dC_t + dI_t$. Dividing by $\bar{Y}$ we get: $\frac{dY_t}{\bar{Y}} = \frac{1}{\bar{Y}} dC_t + \frac{1}{\bar{Y}} dI_t$. Then: $\frac{dY_t}{\bar{Y}} = \frac{\bar{C}}{\bar{Y}} \frac{dC_t}{\bar{C}} + \frac{\bar{I}}{\bar{Y}} \frac{dI_t}{\bar{I}} \Rightarrow \hat{Y}_t = \frac{\bar{C}}{\bar{Y}} \hat{C}_t + \frac{\bar{I}}{\bar{Y}} \hat{I}_t \Rightarrow$

$$\bar{Y} \hat{Y}_t = \bar{C} \hat{C}_t + \bar{I} \hat{I}_t$$

From $\frac{aC_t}{1-H_t} = w_t$ we get: $\ln a + \ln C_t - \ln(1 - H_t) = \ln w_t$. Differentiating we get: $\frac{dC_t}{C_t} + \frac{1}{1-H_t} dH_t = \frac{dw_t}{w_t} \Rightarrow \frac{dC_t}{C_t} + \frac{\bar{H}}{1-\bar{H}} \frac{dH_t}{\bar{H}} = \frac{dw_t}{w_t} \Rightarrow$

$$\hat{C}_t + \frac{\bar{H}}{1-\bar{H}} \hat{H}_t = \hat{w}_t$$

From $\frac{1}{C_t} = \mathbb{E}_t \left\{ \frac{\beta}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$ we get: $-\ln C_t = \ln \mathbb{E}_t \left\{ \frac{\beta}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$. Approximating the log of the expectation with the expectation of the log we get: $-\ln C_t \cong \mathbb{E}_t \{ \ln \beta - \ln C_{t+1} + \ln(r_{t+1} + 1 - \delta) \}$. Differentiating (and dropping the approximation sign): $-\frac{dC_t}{C_t} = \mathbb{E}_t \left\{ -\frac{dC_{t+1}}{C_{t+1}} + \frac{1}{r_{t+1} + 1 - \delta} dr_{t+1} \right\}$. Since $\frac{1}{r_{t+1} + 1 - \delta} = \beta$ we get: $-\frac{dC_t}{C_t} = \mathbb{E}_t \left\{ -\frac{dC_{t+1}}{C_{t+1}} + \beta dr_{t+1} \right\} \Rightarrow -\frac{dC_t}{C_t} = \mathbb{E}_t \left\{ -\frac{dC_{t+1}}{C_{t+1}} + \beta \bar{r} \frac{dr_{t+1}}{\bar{r}} \right\} \Rightarrow$

$$-\hat{C}_t = \mathbb{E}_t \{ -\hat{C}_{t+1} + \beta \bar{r} \hat{r}_{t+1} \} \Rightarrow$$

$$\mathbb{E}_t \hat{C}_{t+1} - \beta \bar{r} \mathbb{E}_t \hat{r}_{t+1} = \hat{C}_t$$

Uhlig's Method

Solving for P using Generalized Eigenvalues and Eigenvectors

We want to find the matrix P that solves:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P + (H - KC^{-1}B) = \mathbf{0}$$

Define:

$$\Psi \equiv F - JC^{-1}A$$

$$\Gamma \equiv JC^{-1}B - G + KC^{-1}A$$

$$\Theta \equiv H - KC^{-1}B$$

Then, we can rewrite our original matrix equation as follows:

$$\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$$

Remark. P, Ψ, Γ, Θ are $m \times m$ matrices, where m is the number of endogenous state variables (*i.e.*, the number of elements in vector x_t).

It will be useful to define the following two matrices:

$$\Xi \equiv \begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix}$$

$$\Delta \equiv \begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix}$$

Remark. In the expressions above, I denotes the $m \times m$ identity matrix, and $\mathbf{0}$ denotes an $m \times m$ matrix of zeros. Hence, Ξ and Δ are $2m \times 2m$ matrices.

Definition: Generalized Eigenvalue Problem.

The generalized eigenvalue problem for a pair of (possibly complex) $n \times n$ matrices (Ξ, Δ) is the problem of finding the (possibly complex) scalar λ and the (possibly complex) $n \times 1$ vectors $v \neq \mathbf{0}$ that satisfy $\Xi v = \lambda \Delta v$. The values of λ that satisfy the equation are the *generalized eigenvalues* (of Ξ with respect to Δ), and the corresponding vectors $v \neq \mathbf{0}$ are the *generalized (right) eigenvectors*.

Remark 1. When Δ is nonsingular we can reduce the generalized eigenvalue problem to a standard eigenvalue problem. Premultiplying $\Xi x = \lambda \Delta x$ by Δ^{-1} we get $\Delta^{-1}\Xi x = \lambda x$. Hence, the generalized eigenvalues of Ξ with respect to Δ coincide with the standard eigenvalues of $\Delta^{-1}\Xi$.

Remark 2. In our particular case, $n = 2m$. Hence, there are $2m$ generalized eigenvalues (but some may be repeated).

Claim 1 (Theorem 3.4, Part 1, in Uhlig (1999))

Let v be a generalized eigenvector of Ξ with respect to Δ , and λ its associated generalized eigenvalue. Then, v can be written in the following way:

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda x \\ x \end{bmatrix} \text{ for some } x \text{ of dimension } m \times 1.$$

Proof

From $\Xi v = \lambda \Delta v$ we get:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Then:

$$(1) \quad \Gamma v_1 + \Theta v_2 = \lambda \Psi v_1$$

$$(2) \quad v_1 = \lambda v_2$$

From (2):

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda v_2 \\ v_2 \end{bmatrix}$$

Finally, setting $x = v_2$:

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda x \\ x \end{bmatrix}$$

■

Claim 2 (Theorem 3.4, Part 2, in Uhlig (1999))

Suppose there are m generalized eigenvalues of Ξ w.r.t. Δ - $\lambda_1, \dots, \lambda_m$ - such that their associated generalized eigenvectors - $v_1, \dots, v_m$ - can be written as $v_1 = \begin{bmatrix} \lambda_1 x_1 \\ x_1 \end{bmatrix}, \dots, v_m = \begin{bmatrix} \lambda_m x_m \\ x_m \end{bmatrix}$ with $x_1, \dots, x_m$ linearly independent. Then

$$P = \Omega \Lambda \Omega^{-1}$$

is a solution to $\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$, where $\Omega \equiv [x_1, x_2, \dots, x_m]$ and $\Lambda \equiv \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$.

Proof.

First notice that we can always write $v_i = \begin{bmatrix} \lambda_i x_i \\ x_i \end{bmatrix}$ because of Claim 1. Notice also that Ω and Λ are $m \times m$ matrices. Moreover, Ω is invertible because of the assumption that $x_1, \dots, x_m$ are linearly independent.

From (1) above we get: $\Gamma v_1 + \Theta v_2 = \lambda \Psi v_1$. Since $v = \begin{bmatrix} \lambda x \\ x \end{bmatrix}$ we get: $\Gamma \lambda x + \Theta x = \lambda \Psi \lambda x$. Then:

$$\Psi x \lambda^2 - \Gamma x \lambda - \Theta x = \mathbf{0}$$

Applying the expression above to our m eigenvalues/eigenvectors we get:

$$\begin{aligned}\Psi x_1 \lambda_1^2 - \Gamma x_1 \lambda_1 - \Theta x_1 &= \mathbf{0} \\ \Psi x_m \lambda_m^2 - \Gamma x_m \lambda_m - \Theta x_m &= \mathbf{0}\end{aligned}$$

Recall that the left-hand and right-hand sides of the expressions above are $m \times 1$ matrices. We can put them all together in the following $m \times m$ matrices:

$$[\Psi x_1 \lambda_1^2 - \Gamma x_1 \lambda_1 - \Theta x_1, \dots, \Psi x_m \lambda_m^2 - \Gamma x_m \lambda_m - \Theta x_m] = [\mathbf{0}, \dots, \mathbf{0}]$$

Now we can write:

$$[\Psi x_1 \lambda_1^2, \dots, \Psi x_m \lambda_m^2] - [\Gamma x_1 \lambda_1, \dots, \Gamma x_m \lambda_m] - [\Theta x_1, \dots, \Theta x_m] = \mathbf{0}$$

where the right-hand side is an $m \times m$ null matrix.

Then:

$$\Psi [x_1 \lambda_1^2, \dots, x_m \lambda_m^2] - \Gamma [x_1 \lambda_1, \dots, x_m \lambda_m] - \Theta [x_1, \dots, x_m] = \mathbf{0}$$

Then:

$$\Psi [x_1, x_2, \dots, x_m] \begin{bmatrix} \lambda_1^2 & 0 & \dots & 0 \\ 0 & \lambda_2^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_m^2 \end{bmatrix} - \Gamma [x_1, x_2, \dots, x_m] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_m \end{bmatrix} - \Theta [x_1, x_2, \dots, x_m] = \mathbf{0}$$

Then:

$$\Psi \Omega \Lambda^2 - \Gamma \Omega \Lambda - \Theta \Omega = \mathbf{0}$$

Since Ω is invertible we can post-multiply the expression above by Ω^{-1} . We get: $\Psi \Omega \Lambda^2 \Omega^{-1} - \Gamma \Omega \Lambda \Omega^{-1} - \Theta \Omega \Omega^{-1} = \mathbf{0} \Omega^{-1} \Rightarrow$

$$(3) \quad \Psi \Omega \Lambda^2 \Omega^{-1} - \Gamma \Omega \Lambda \Omega^{-1} - \Theta = \mathbf{0}$$

Now define

$$P \equiv \Omega \Lambda \Omega^{-1}$$

and notice that $P^2 = \Omega \Lambda \Omega^{-1} \Omega \Lambda \Omega^{-1} = \Omega \Lambda \Lambda \Omega^{-1} \Rightarrow$

$$P^2 = \Omega \Lambda^2 \Omega^{-1}$$

Then, we can rewrite (3) as follows:

$$\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$$

which is what we wanted to show.

■

Stability

Consider a process of the form:

$$x_t = \rho x_{t-1} + \epsilon_t$$

where x_t and ϵ_t are scalar random variables. Assume ϵ_t is stationary. It is well known that the stability condition for such a process is $|\rho| < 1$.¹

Now consider the following generalization of the expression above:

$$x_t = Px_{t-1} + Qz_t$$

where x_t and z_t are vectors, P and Q are matrices, and z_t is stationary. In this case, stability requires that each eigenvalue of P has modulus less than one ("it lies within the unit circle"). The matrix Q does not play any role because z_t has been assumed to be stationary.

Now recall from Claim 2 that P can be written as follows: $P = \Omega\Lambda\Omega^{-1}$. From this, it follows that the eigenvalues of P coincide with the eigenvalues of Λ : $Px = \lambda x \Rightarrow \Omega\Lambda\Omega^{-1}x = \lambda x \Rightarrow \Omega^{-1}\Omega\Lambda\Omega^{-1}x = \lambda\Omega^{-1}x \Rightarrow \Lambda\Omega^{-1}x = \lambda\Omega^{-1}x \Rightarrow \Lambda v = \lambda v$, where $v = \Omega^{-1}x$ is an eigenvector of Λ associated to λ . Since Λ is diagonal, its eigenvalues coincide with its diagonal elements. Now, according to Claim 2, the diagonal elements of Λ are m of the generalized eigenvalues of Ξ w.r.t. Δ . Therefore, we conclude that our recursive solution $x_t = Px_{t-1} + Qz_t$ will be stable as long as we can find m generalized eigenvalues of Ξ w.r.t. Δ with modulus lower than one.

If there are exactly m generalized eigenvalues with modulus less than one, there will be a unique stable solution (there may well be additional unstable solutions). Notice that m is the dimension of the vector x_t . Therefore, the stable solution will be unique when the number of eigenvalues with modulus less than one coincides with the number of endogenous state variables.

If there are more than m eigenvalues with modulus less than one, then there will be indeterminacy, *i.e.*, multiple stable solutions (we can construct multiple P s with eigenvalues lower than one in modulus).

If there are less than m eigenvalues with modulus less than one there will be no stable solution.

¹Loosely speaking, stability means that x_t does not explode over time. Iterating over $x_t = \rho x_{t-1} + \epsilon_t$ we get $x_t = \rho^{n+1}x_{t-(n+1)} + \rho^n\epsilon_{t-n} + \dots + \rho\epsilon_{t-1} + \epsilon_t = \rho^{n+1}y_{t-(n+1)} + \sum_{j=0}^n \rho^j \epsilon_{t-j}$. When $|\rho| < 1$, $\lim_{n \rightarrow \infty} \rho^{n+1} = 0$. Moreover, for a stationary solution we have: (i) $\lim_{n \rightarrow \infty} \rho^{n+1}x_{t-(n+1)} = 0$ (in an appropriate sense); and (ii) $\lim_{n \rightarrow \infty} \sum_{j=0}^n \rho^j \epsilon_{t-j} = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-j}$ is well defined. Therefore, we get: $x_t = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-j}$.

Uhlig's Method

Solving for P using the Generalized Schur Decomposition

We want to find the matrix P that solves:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P + (H - KC^{-1}B) = \mathbf{0}$$

Define:

$$\Psi \equiv F - JC^{-1}A$$

$$\Gamma \equiv JC^{-1}B - G + KC^{-1}A$$

$$\Theta \equiv H - KC^{-1}B$$

Then, we can rewrite our original matrix equation as follows:

$$\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$$

Remark. P, Ψ, Γ, Θ are $m \times m$ matrices, where m is the number of endogenous state variables (*i.e.*, the number of elements in vector x_t).

It will be useful to define the following two matrices:

$$\Xi \equiv \begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix}$$
$$\Delta \equiv \begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix}$$

Remark. In the expressions above, I denotes the $m \times m$ identity matrix, and $\mathbf{0}$ denotes an $m \times m$ matrix of zeros. Hence, Ξ and Δ are $2m \times 2m$ matrices.

Now we are going to apply the Generalized Schur Decomposition to the matrix pair (Ξ, Δ) . We start by recalling the notion of a generalized eigenvalue of a matrix pair, and then we state the Generalized Schur Decomposition Theorem.

Definition: Generalized Eigenvalue Problem.

The generalized eigenvalue problem for a pair of (possibly complex) $n \times n$ matrices (Ξ, Δ) is the problem of finding the (possibly complex) scalar λ and the (possibly complex) $n \times 1$ vectors $v \neq \mathbf{0}$ that satisfy $\Xi v = \lambda \Delta v$. The values of λ that satisfy the equation are the *generalized eigenvalues* (of Ξ with respect to Δ), and the corresponding vectors $v \neq \mathbf{0}$ are the *generalized (right) eigenvectors*.

Remark. When Δ is nonsingular we can reduce the generalized eigenvalue problem to a standard eigenvalue problem. Premultiplying $\Xi x = \lambda \Delta x$ by Δ^{-1} we get $\Delta^{-1}\Xi x = \lambda x$. Hence, the generalized eigenvalues of Ξ with respect to Δ coincide with the standard eigenvalues of $\Delta^{-1}\Xi$.

Theorem: Generalized Schur Decomposition.

Let Ξ and Δ be (possibly) complex $n \times n$ matrices. Then, there exist (possibly) complex, unitary $n \times n$ matrices Q and Z such that

$$\begin{aligned}\Xi &= QTZ \\ \Delta &= QSZ\end{aligned}$$

where S and T are (possibly) complex, upper triangular $n \times n$ matrices. The diagonal elements of T divided by the diagonal elements of S , $\frac{t_{ii}}{s_{ii}}$, are the generalized eigenvalues of the matrix pair (Ξ, Δ) . If $s_{ii} \neq 0$ and $t_{ii} \neq 0$, the generalized eigenvalue is finite; if $s_{ii} = 0$ and $t_{ii} \neq 0$, the generalized eigenvalue is infinite (by convention); if $s_{ii} = 0$ and $t_{ii} = 0$, the matrix pair (Ξ, Δ) is singular (*i.e.*, $|\Xi - \lambda\Delta| = 0$) and there is an infinity of generalized eigenvalues. Furthermore, the pairs (s_{ii}, t_{ii}) can be arranged in any order.

Remark 1. In our particular case, $n = 2m$.

Remark 2. A matrix Z is unitary if and only if $Z'Z = I$, where Z' denotes the conjugate transpose of Z , and I is the identity matrix. The conjugate transpose of Z is formed by transposing Z and replacing each element with its complex conjugate.

Remark 3. A maxtrix S is upper triangular when each element below the diagonal is equal to zero.

Remark 4. The decomposition above is usually called the QZ decomposition, in reference to the two unitary matrices mentioned in the theorem.

Remark 5. A QZ decomposition always exists, but it may not be unique.

Remark 6. Do not confuse matrices Q and S in the theorem above with the matrices Q and S that we used in our recursive representation of the solution ($x_t = Px_{t-1} + Qz_t$, $y_t = Rx_{t-1} + Sz_t$).

Now we partition Q , Z , S , and T comformably with Ξ and Δ . We get (recall that S and T are upper triangular):

$$\begin{aligned}Q &= \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix}, Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \\ S &= \begin{bmatrix} S_{11} & S_{12} \\ \mathbf{0} & S_{22} \end{bmatrix}, T = \begin{bmatrix} T_{11} & T_{12} \\ \mathbf{0} & T_{22} \end{bmatrix}\end{aligned}$$

where each block is an $m \times m$ matrix.

Claim (Theorem 3.5, Part 1, in Uhlig (1999))

Suppose Z_{21} and Q_{21} are nonsingular (*i.e.*, invertible). Then

$$P = -Z_{21}^{-1}Z_{22}$$

is a solution to $\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$.

Proof

To show that $P = -Z_{21}^{-1}Z_{22}$ solves $\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$ we can show that

$$P = -Z_{21}^{-1}Z_{22} \quad \Rightarrow \quad (\Psi P^2 - \Gamma P - \Theta)x = \mathbf{0} \quad \forall x \in \mathbb{R}^m$$

From $(\Psi P^2 - \Gamma P - \Theta)x = \mathbf{0}$ we get:

$$\Psi P^2 x = \Gamma P x + \Theta x$$

We also have (trivially):

$$Px = Px$$

Combining the expressions above we get:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2 x \\ Px \end{bmatrix} = \begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix}$$

More compactly, we can write the expression above as follows:

$$\Delta \begin{bmatrix} P^2 x \\ Px \end{bmatrix} = \Xi \begin{bmatrix} Px \\ x \end{bmatrix}$$

Therefore, we want to show that

$$P = -Z_{21}^{-1}Z_{22} \quad \Rightarrow \quad \Delta \begin{bmatrix} P^2 x \\ Px \end{bmatrix} = \Xi \begin{bmatrix} Px \\ x \end{bmatrix} \quad \forall x \in \mathbb{R}^m$$

Notice that the second block in the system above must hold for sure (since it is just $Px = Px$). On the other hand, the first block of the system is what we really need to show (*i.e.*, that $\Psi P^2 x = \Gamma P x + \Theta x$). To prove this we'll do the following: (i) Fix an arbitrary $x \in \mathbb{R}^m$; (ii) rewrite the left-hand side of the system above using $\Delta = QSZ$, partition conformably, and assume $P = -Z_{21}^{-1}Z_{22}$; (iii) rewrite the right-hand side using $\Xi = QTZ$, partition conformably, and assume $P = -Z_{21}^{-1}Z_{22}$; (iv) impose equality between the second block of the left-hand side and the second block of the right-hand side, and show that this implies equality between the first block in the left-hand side and the first block in the right-hand side.

For any $x \in \mathbb{R}^m$, we can use the QZ decomposition to rewrite the left-hand side as follows:

$$\Delta \begin{bmatrix} P^2x \\ Px \end{bmatrix} = QSZ \begin{bmatrix} P^2x \\ Px \end{bmatrix}$$

Partitioning the matrices above we get:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ \mathbf{0} & S_{22} \end{bmatrix} \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11} & Q_{11}S_{12} + Q_{12}S_{22} \\ Q_{21}S_{11} & Q_{21}S_{12} + Q_{22}S_{22} \end{bmatrix} \begin{bmatrix} Z_{11}P^2x + Z_{12}Px \\ Z_{21}P^2x + Z_{22}Px \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11} & Q_{11}S_{12} + Q_{12}S_{22} \\ Q_{21}S_{11} & Q_{21}S_{12} + Q_{22}S_{22} \end{bmatrix} \begin{bmatrix} Z_{11}P^2x + Z_{12}Px \\ (Z_{21}P + Z_{22})Px \end{bmatrix}$$

If $P = -Z_{21}^{-1}Z_{22}$ we get: $Z_{21}P + Z_{22} = -Z_{21}Z_{21}^{-1}Z_{22} + Z_{22} = -Z_{22} + Z_{22} = \mathbf{0}$.

Then:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11} & Q_{11}S_{12} + Q_{12}S_{22} \\ Q_{21}S_{11} & Q_{21}S_{12} + Q_{22}S_{22} \end{bmatrix} \begin{bmatrix} Z_{11}P^2x + Z_{12}Px \\ \mathbf{0} \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Psi P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11}(Z_{11}P^2x + Z_{12}Px) \\ Q_{21}S_{11}(Z_{11}P^2x + Z_{12}Px) \end{bmatrix}$$

Then:

$$(1) \quad \Psi P^2x = Q_{11}S_{11}(Z_{11}P^2x + Z_{12}Px)$$

$$(2) \quad Px = Q_{21}S_{11}(Z_{11}P^2x + Z_{12}Px)$$

Analogously, for any $x \in \mathbb{R}^m$, we can use the QZ decomposition to rewrite the right-hand side as follows:

$$\Xi \begin{bmatrix} Px \\ x \end{bmatrix} = QTZ \begin{bmatrix} Px \\ x \end{bmatrix}$$

Partitioning the matrices above we get:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} \begin{bmatrix} T_{11} & T_{12} \\ \mathbf{0} & T_{22} \end{bmatrix} \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11} & Q_{11}T_{12} + Q_{12}T_{22} \\ Q_{21}T_{11} & Q_{21}T_{12} + Q_{22}T_{22} \end{bmatrix} \begin{bmatrix} Z_{11}Px + Z_{12}x \\ Z_{21}Px + Z_{22}x \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11} & Q_{11}T_{12} + Q_{12}T_{22} \\ Q_{21}T_{11} & Q_{21}T_{12} + Q_{22}T_{22} \end{bmatrix} \begin{bmatrix} Z_{11}Px + Z_{12}x \\ (Z_{21}P + Z_{22})x \end{bmatrix}$$

If $P = -Z_{21}^{-1}Z_{22}$ we get: $Z_{21}P + Z_{22} = -Z_{21}Z_{21}^{-1}Z_{22} + Z_{22} = -Z_{22} + Z_{22} = \mathbf{0}$.

Then:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11} & Q_{11}T_{12} + Q_{12}T_{22} \\ Q_{21}T_{11} & Q_{21}T_{12} + Q_{22}T_{22} \end{bmatrix} \begin{bmatrix} Z_{11}Px + Z_{12}x \\ \mathbf{0} \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Gamma Px + \Theta x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11}(Z_{11}Px + Z_{12}x) \\ Q_{21}T_{11}(Z_{11}Px + Z_{12}x) \end{bmatrix}$$

Then:

$$(3) \quad \Gamma Px + \Theta x = Q_{11}T_{11}(Z_{11}Px + Z_{12}x)$$

$$(4) \quad Px = Q_{21}T_{11}(Z_{11}Px + Z_{12}x)$$

We know (2) and (4) must be equal to each other. Hence, $Q_{21}S_{11}(Z_{11}P^2x + Z_{12}Px) = Q_{21}T_{11}(Z_{11}Px + Z_{12}x)$. Since, by assumption, Q_{21} is invertible, we can premultiply by Q_{21}^{-1} to get:

$$(5) \quad S_{11}(Z_{11}P^2x + Z_{12}Px) = T_{11}(Z_{11}Px + Z_{12}x)$$

Substituting (5) into (3) we get:

$$(6) \quad \Gamma Px + \Theta x = Q_{11}S_{11}(Z_{11}P^2x + Z_{12}Px)$$

But the right-hand side of (6) coincides with the right hand side of (1). Therefore, the left-hand sides must coincide too. Hence, we have shown that:

$$(7) \quad P = -Z_{21}^{-1}Z_{22} \Rightarrow \Psi P^2x = \Gamma Px + \Theta x$$

or, equivalently,

$$(8) \quad P = -Z_{21}^{-1}Z_{22} \Rightarrow (\Psi P^2 - \Gamma P - \Theta)x = \mathbf{0}$$

Since $x \in \mathbb{R}^m$ was chosen arbitrarily, we conclude that the implication above holds $\forall x \in \mathbb{R}^m$, which is what we wanted to show. ■